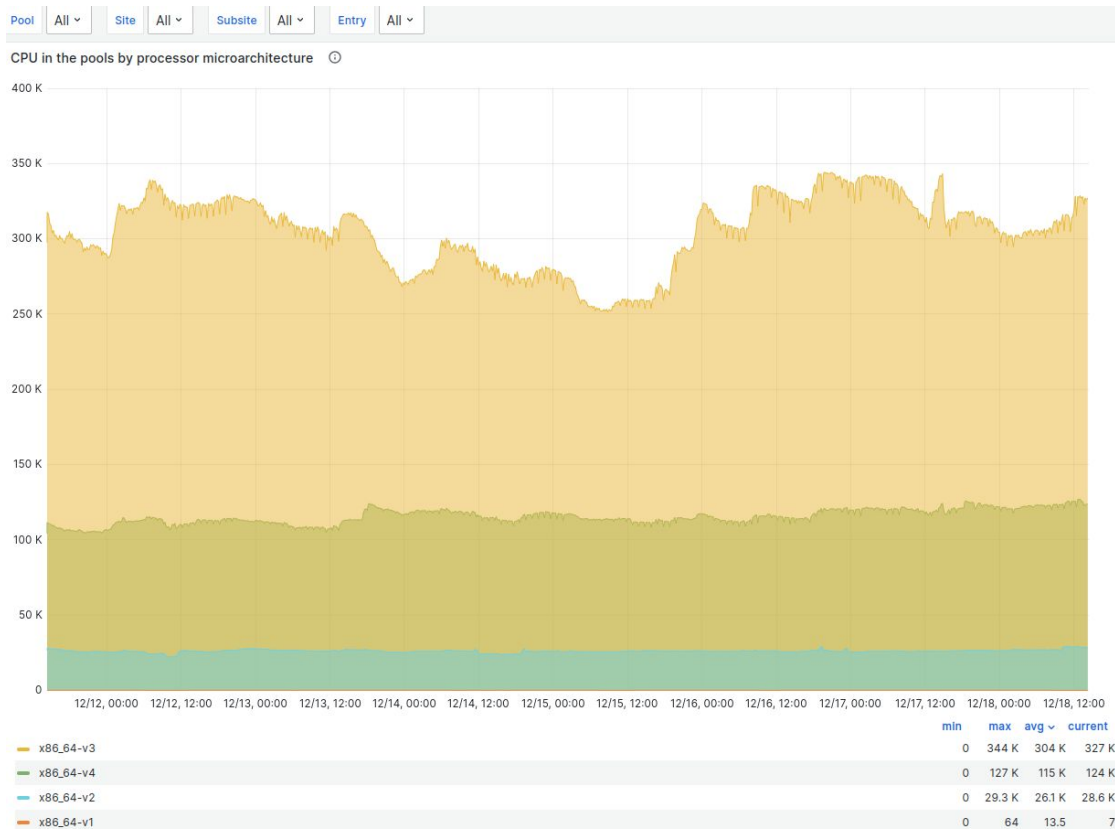


Plan to support various microarch in production

Marco Mascheroni on behalf of the Submission Infrastructure
and DMWM/CRAB Teams

Pool resources by microarchs => no v1

- Most resources are already v3 and v4
- Pilots starting on v1 resources fails in validation sted
 - Uses `/lib64/ld-linux-x86-64.so.2` with fallback to `lscpu`



New requirement from last O&C week

- **Starting with CMSSW_15_0_0_pre1**, we switch from v2 to v3 for validation and **maintain validated v3 versions** moving forward.
 - a. The High-Level Trigger (**HLT**) will benefit from this by **not needing to perform separate validation for v3**, simplifying the validation process and reducing redundancy.
- Need to implement a **matching condition**: jobs submitted with **CMSSW_15_0_0** (and later) are directed to sites that support **v3-only CPUs**.
 - a. Sites restricted to v2 CPUs will continue to receive jobs for Run-2 and Run-3 campaigns
 - b. This ensures that all sites remain actively engaged in ongoing data processing without requiring immediate hardware upgrades.
- **SCRAM sets an environment variable** to expose default micro-arch of a release
 - a. Central Production/RelVals will use that in the job requirement (scheduled for next quarter).
 - b. Analysis users can overwrite it if they want to

Implementation status

- Condor matching condition implemented and available in Integration Testbed
 - `Microarch` is undefined || `REQUIRED_MINIMUM_MICROARCH` is undefined ||
`REQUIRED_MINIMUM_MICROARCH <= int(substr(split(Microarch, "-") [1], 1))`
 - Protections for undefined `Microarch` (on aarch64 machines) and undefined requirement
 - SI implementation could be pushed it in production before the break if needed
- WMAgent and CRAB can set a **requirement** like:
 - `REQUIRED_MINIMUM_MICROARCH = 3`
- CRAB implementation already available through `crab-dev`
 - <https://github.com/dmwm/CRABClient/pull/5362>
 - <https://github.com/dmwm/CRABServer/pull/8866>
 - <https://github.com/dmwm/CRABServer/pull/8860>

Relvals for CMSSW_15_0_0

- Minimum amount of v2 resources at FNAL
- Could Relval be executed at FNAL without final implementation?

